Aligning objectives, endpoints and statistical analyses

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Life cycle for a clinical trial

Plan → Start → Conduct → End → Analyze → Report

Statisticians are involved in all stages of the life cycle and member of the study team

Estimand framework

Trial objective

- Research question

Estimand

- Description of the treatment effect

Estimator

- Analysis method to compute an estimate of the estimand

Estimate

- A numerical value computed by an estimator

Planning – first step – define the research question - example

- To evaluate the efficacy of dupilumab compared to placebo in patients with uncontrolled persistent asthma.

Planning – next step – define the estimand

- Patient population
- Treatment and control group
- Endpoint / Outcome
- Population-level summary
- Intercurrent event strategy

Example of an estimand

"The estimand is the difference in the mean FEV1 at Month 2 between dupilumab versus placebo, in patients with uncontrolled persistent asthma, if they were to continue using dupilumab over the entire trial period without the use of rescue treatment."

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Patient population

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Treatment and control group

Example of an estimand

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Endpoint / Outcome including time

Example of an estimand

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Population-level summary

Example of an estimand

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Intercurrent event strategy

Intercurrent events are

- Post randomisation occurrences that affect either the interpretation or the existence of the endpoint/outcome
  - Stopping the assigned treatment early
  - Not starting the assigned treatment
  - Receiving treatment not according to randomisation
  - Receiving “forbidden” medication (eg, rescue medication)
  - Death

Intercurrent event strategy - example

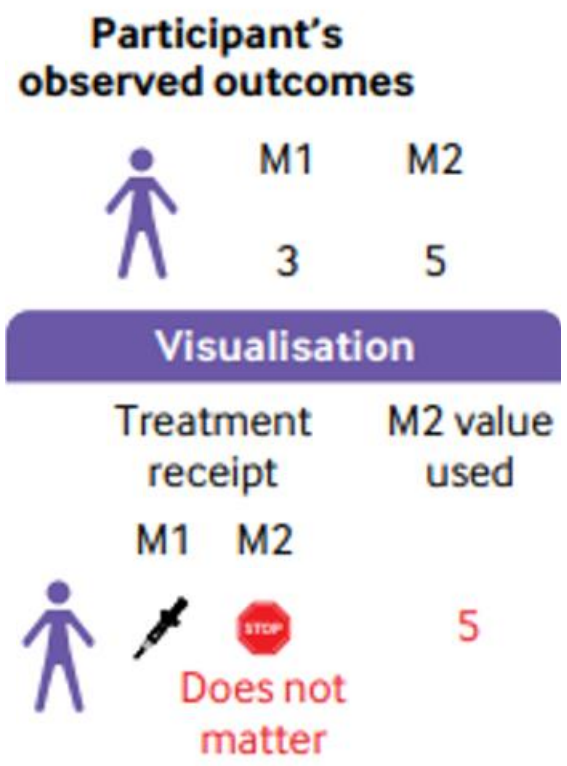
- A randomised trial compares intervention with control to understand how outcomes differ at month 2
- However, one participant stops treatment before month 2

Participant's observed outcomes

	M1	M2
	3	5

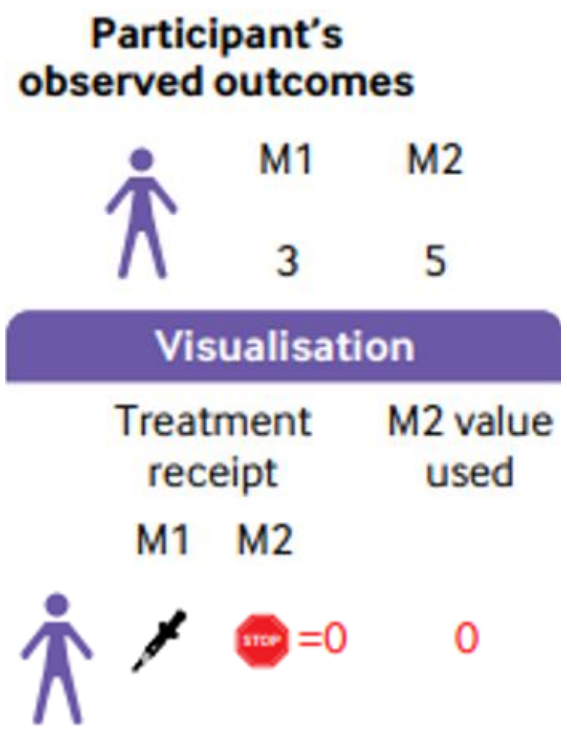
Treatment policy

- Intercurrent event is considered irrelevant
- Use outcome data regardless of intercurrent events.



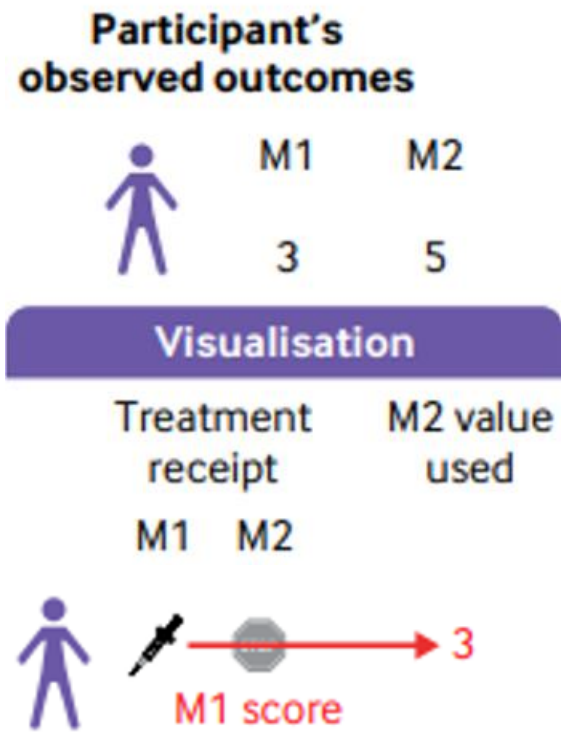
Composite

- An intercurrent event is considered informative in itself
- Set outcomes to “failure”



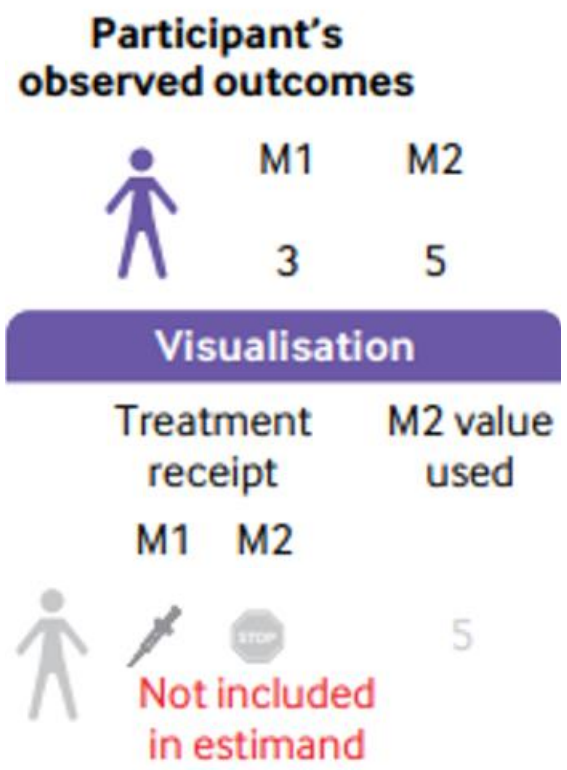
While on treatment

- Outcome up to the point of the event are relevant, but outcome data after the event are not
- Change the outcome for patients who had the event to only include outcome data before the event



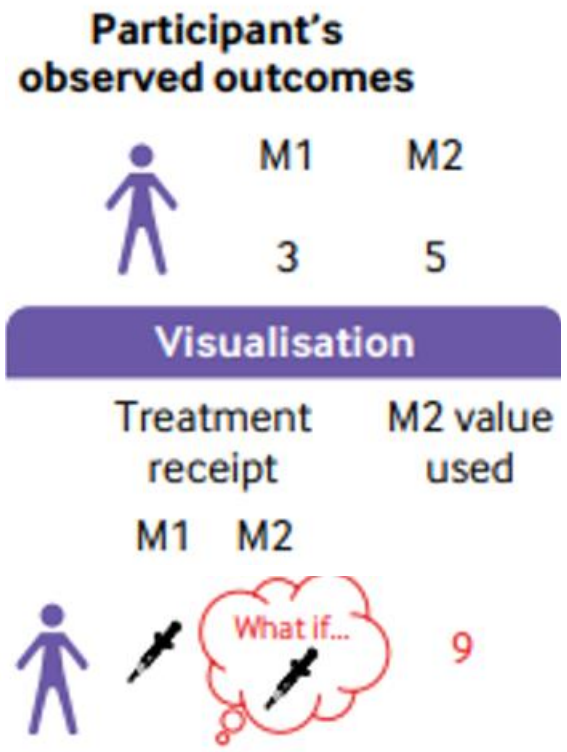
- Principal stratum

- Outcome data for patients with the event are not relevant
- Exclude patients who experience the event in the analysis population



Hypothetical

- Outcome data after the event are not relevant
- What if?
- Use a statistical model to estimate what patient outcomes would have been if the intercurrent event did not occur



Example of an estimand

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Intercurrent event strategy: hypothetical

Choice of statistical analysis method (estimator) is driven by the endpoint and population-level summary – some examples

Endpoint	Population-level summary	Estimator Analysis method to compute an estimate of the estimand
FEV_1 at Month 2 (continuous)	Mean difference between treatment groups	Mixed Model for Repeated Measures estimating least-squares mean difference
Pain score at Week 12 on a 0–10 scale (continuous)	Adjusted mean score difference	ANCOVA model with baseline score as covariate
Proportion of patients achieving ≥75% improvement at Week 16 (binary)	Risk difference (or risk ratio / odds ratio)	Logistic regression model adjusting for baseline covariates
Time to first cardiovascular event (time-to-event)	Hazard ratio	Cox proportional hazards model
Overall survival at 24 months (time-to-event)	Survival probability difference at 24 months	Kaplan-Meier survival estimate

Final remarks

- Estimand framework is recommended by the regulatory authorities and is a tool to design a successful clinical trial
- Keep the patients in the study
- Plan sensitivity analyses to explore robustness of the conclusions from the primary statistical analysis

Read more about estimand framework

→ ICH E9 (R1) addendum on estimands and sensitivity analysis in clinical trials to the guideline on statistical principles for clinical trials

→ Kahan and Devane "Estimands: what they are and why we should use them" Journal of Clinical Epidemiology, 2025

Questions?

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Participant's observed outcomes

M1	M2
3	5

Strategy	Explanation	Visualisation	
		Treatment receipt (M1 M2)	M2 value used
Treatment policy Use outcome regardless of intercurrent event	Participant's outcome was 5, so this value is used regardless of fact that they experienced an intercurrent event	Does not matter	5
Composite Assign particular outcome value to those experiencing an intercurrent event	Investigators decided that those who experienced an intercurrent event would be assigned an outcome value of 0	=0	0
While on treatment Use outcome(s) before an intercurrent event occurence	Because the intercurrent event occurred before their M2 score, participant's M1 score of 3 is used in its place	M1 score	3
Hypothetical Consider what outcome would have been if intercurrent event had not occurred	If participant had continued treatment to M2, their outcome would have been 9. However, this value is unknown in practice so must be estimated	What if...	9
Prinicpal stratum Restrict population to those who would not experience an intercurrent event	Because of intercurrent event occurrence, participant is not included in estimand population	Not included in estimand	5